HW7

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NOTE: I have a Mac so all SAS is run on SAS Studio online. So, I will copy any code I use into a code block here, but all outputs I will need to provide screenshots/regenerate them in R.

1

Y and e are written with subscripts of block,treatment, ie y_{12} is block 1 treatment 2.

$$Y = \begin{pmatrix} y_{11} \\ y_{12} \\ y_{21} \\ y_{22} \\ y_{31} \\ y_{32} \end{pmatrix}$$

X is n by num_treatments. Which 'treatment' is each response from?

$$X = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\beta = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$$

Z is n by num_blocks. Which 'block' is each response from?

$$Z = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$u = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}$$

$$e = \begin{pmatrix} e_{11} \\ e_{12} \\ e_{21} \\ e_{22} \\ e_{31} \\ e_{32} \end{pmatrix}$$

2

```
set.seed(1337)
treat_means <- c(100, 110)
sig2_e <- 16
sig2_b <- 12
n_treat <- 2
n_block <- 10
n_samp <- 5
u <- rnorm(n_block, 0, sqrt(sig2_b))

gen_data <- function() {
    treatment <- factor(paste0("Trt_", 1:n_treat))
    block <- factor(paste0("Blk_", 1:n_block))
    reps <- 1:n_samp

    design_matrix <- expand.grid(Rep = reps, Treatment = treatment, Block = block)

    y <- unlist(lapply(1:n_treat, \(i) lapply(1:n_block, \(j) lapply(1:n_samp, \(k) treat_mea
    dat <- as.data.frame(cbind(y, design_matrix$Block, design_matrix$Treatment))
    colnames(dat) <- c("y", "Block", "Treatment")
    dat$Block <- as.factor(dat$Block)
    dat$Treatment <- as.factor(dat$Treatment)
```

```
    dat
}

dt <- gen_data()
write.csv(dt, 'q2_gen_data.csv', row.names = FALSE)
```

Not going to show the generated data, but I export it to CSV to get into SAS Studio.

3

```
PROC IMPORT DATAFILE='/home/u49685591/STAT537/HW7/q2_gen_data.csv'
    DBMS=CSV
    OUT=WORK.HW7
    REPLACE;
    GETNAMES=YES;
RUN;

proc mixed data=HW7 method = REML;
class Block Treatment;
model y = Treatment / s ddfm=satterth;
random Block;
run;
```

```
sas_sigma2_b <- 30.4196
sas_sigma2_e <- 27.3545
sas_trt_diff <- -1.146
```

The estimated output from SAS indicates a σ_b^2 of 30.4196 and a σ_e^2 of 27.3545, which are nowhere near the actual values provided. It also estimated that the difference in means is -1.146, which does show Treatment 1 being lower than Treatment 2, but not by much.

4

```
sas_fixed_f_stat <- 1.2
sas_fixed_p_val <- .2762
```

The F Statistic provided by the output of SAS for the test of whether there is a significant treatment effect is 1.2. This gave a P-value of 0.2762 which implies this is not significant.

5

```
suppressPackageStartupMessages(library('lme4'))
mod <- lmer(y ~ Treatment + (1|Block), data = dt)
summary(mod)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: y ~ Treatment + (1 | Block)
   Data: dt

REML criterion at convergence: 632.7

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.67991 -0.51338  0.07664  0.65787  2.05296

Random effects:
 Groups   Name        Variance Std.Dev.
 Block    (Intercept) 30.42    5.515
 Residual             27.35    5.230
Number of obs: 100, groups:  Block, 10

Fixed effects:
            Estimate Std. Error t value
(Intercept)  107.165      1.894  56.567
Treatment2     1.146      1.046   1.096

Correlation of Fixed Effects:
           (Intr)
Treatment2 -0.276
```

```
# Outputs from Summary
r_lmer_sigma2_b <- 30.42
r_lmer_sigma2_e <- 27.35
r_lmer_trt1_mean <- 107.165 # Intercept of the fixed effects
r_lmer_trt2_mean <- 105.704 + 1.146
```

We got the exact same values (within rounding) as the output from SAS.

The estimated output from R indicates a σ_b^2 of 30.42 and a σ_e^2 of 27.35, which, again, are nowhere near the actual values provided. It also estimated that $\beta_{Trt1} = 107.165$ and β_{Trt2}

= 106.85, which does show Treatment 1 being lower than Treatment 2, but not by much (the typical interpretation of Treatment 2 as a difference of Treatment 1 would be β_{Trt2} = -0.315).

6

```
mod_ml <- lmer(y ~ Treatment + (1|Block), data = dt, REML = FALSE)
summary(mod_ml)
```

```
Linear mixed model fit by maximum likelihood  ['lmerMod']
Formula: y ~ Treatment + (1 | Block)
   Data: dt

      AIC       BIC    logLik -2*log(L)  df.resid
    645.6     656.0    -318.8     637.6        96

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.70364 -0.52539  0.07862  0.66560  2.07168

Random effects:
 Groups   Name        Variance Std.Dev.
 Block    (Intercept) 27.13    5.209
 Residual             27.05    5.201
Number of obs: 100, groups:  Block, 10

Fixed effects:
            Estimate Std. Error t value
(Intercept)  107.165      1.804  59.404
Treatment2     1.146      1.040   1.102

Correlation of Fixed Effects:
           (Intr)
Treatment2 -0.288
```

```
# Outputs from Summary
r_ml_sigma2_b <- 27.13
r_ml_sigma2_e <- 27.05
```

The β estimates actually remained exactly the same. However, the estimates of the variance changed to σ_b^2 of 27.13 and a σ_e^2 of 27.05 which is still not close to the true values, though technically closer.

7

```
suppressPackageStartupMessages(library('lmerTest'))

# I'm using this package I found online to test the likliehood since ANOVA doesn't work with
# Test using the REML method
ranova(mod)
```

```
ANOVA-like table for random-effects: Single term deletions

Model:
y ~ Treatment + (1 | Block)
            npar  logLik    AIC    LRT Df Pr(>Chisq)
<none>         4 -316.33 640.66
(1 | Block)    3 -339.59 685.17 46.511  1   9.11e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Test using the ML method
ranova(mod_ml)
```

```
ANOVA-like table for random-effects: Single term deletions

Model:
y ~ Treatment + (1 | Block)
            npar  logLik    AIC    LRT Df Pr(>Chisq)
<none>         4 -318.78 645.57
(1 | Block)    3 -341.51 689.03 45.463  1  1.556e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Both tests indicate that, with p-values MUCH less than .05, we reject $H_0 : \sigma_b^2 = 0$. I didn't do anything specific to divide by 2, but I'm assuming the ranova() function already does that.

8

```
mod_red <- lmer(y ~ (1 | Block), data = dt, REML = FALSE)
anova(mod_ml, mod_red)
```

```
Data: dt
Models:
mod_red: y ~ (1 | Block)
mod_ml: y ~ Treatment + (1 | Block)
        npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
mod_red    3 644.77 652.59 -319.39    638.77
mod_ml     4 645.57 655.99 -318.78    637.57 1.2057  1     0.2722
```

Comparing the reduced to the full model, we get a χ^2 statistic of ~1.21 with degrees of freedom of 1, and p-value of ~.27 which indicates that we accept the $H_0 : \beta_{Trt1} = \beta_{Trt2}$. The F-test said this was definitely significant, where as this test says it is not significant.

9

```
remove_rows <- sample(1:100, 20, replace = FALSE)
dt_red <- dt[-remove_rows,]
write.csv(dt_red, 'q9_gen_data.csv', row.names = FALSE)
```

```
PROC IMPORT DATAFILE='/home/u49685591/STAT537/HW7/q9_gen_data.csv'
    DBMS=CSV
    OUT=WORK.HW7
    REPLACE;
    GETNAMES=YES;
RUN;

proc mixed data=HW7 method = REML;
class Block Treatment;
model y = Treatment / s ddfm=satterth;
random Block;
run;
```

In the final Fixed Effects test, the DF changed from 89 in the first analysis to 69.1 in this one, which roughly matches the loss of 20 data points.

10

We got the same change of DF as in #9 (89 to 69.1)

11

```
PROC IMPORT DATAFILE='/home/u49685591/STAT537/HW7/q9_gen_data.csv'
    DBMS=CSV
    OUT=WORK.HW7
    REPLACE;
    GETNAMES=YES;
RUN;

proc mixed data=HW7 method = REML;
class Block Treatment;
model y = Treatment / s ddfm=KenwardRoger;
random Block;
run;
```

I noticed that the F statistic changed from 1.2 to 1.91, though the p-value is still large enough that this is not a significant effect.

12

I perform this back on the original dataset, not the one with 20 removed rows.

```
mod_int <- lmer(y ~ Treatment + (1 | Block) + (1 | Treatment:Block), data = dt)
anova(mod, mod_int)
```

```
refitting model(s) with ML (instead of REML)

Data: dt
Models:
mod: y ~ Treatment + (1 | Block)
mod_int: y ~ Treatment + (1 | Block) + (1 | Treatment:Block)
        npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
mod        4 645.57 655.99 -318.78    637.57
mod_int    5 623.73 636.76 -306.86    613.73 23.836  1  1.049e-06 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Even though we know the interaction doesn't exist, we do indeed reject the $H_0 : \sigma^2_{tb} = 0$ and we conclude that there is a significant interaction.